



Modelling Financial Markets

Seminar 1: Stylised Facts and Returns — Hands-on



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARCH-LM: ARCH Lagrange Multiplier (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ EGARCH: Exponential GARCH
- ▣ ETF: Exchange-Traded Fund
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ JB: Jarque–Bera (test)
- ▣ OHLC: Open, High, Low, Close
- ▣ OLS: Ordinary Least Squares
- ▣ QQ: Quantile–Quantile (plot)
- ▣ RMSE: Root Mean Squared Error
- ▣ SE: Standard Error
- ▣ TSPL: Time-Shifted Power Law (kernel)

Acronyms used in this material (2/2)

- **US: United States**
- **VIX: CBOE Volatility Index**

Outline

Introduction

Key Definitions

Part A: Pen-and-Paper Exercises

Part B: Verify (B1–B8)

Part B: Test (B9–B15)

Part C: Open Question

Wrap-up

Seminar Objectives

By the end of this seminar you can

- ▣ Compute and interpret the statistics behind the stylised facts
- ▣ Derive where heavy tails and volatility clustering come from (mixtures, regimes, GARCH)
- ▣ Estimate tail indices and decide which moments exist
- ▣ Test autocorrelation, asymmetry and performance differences with inference robust to heavy tails and clustering
- ▣ Check which facts a model reproduces

▣ Background: Lecture 1 and [Cont \(2001\)](#); textbook: [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eleven pen-and-paper exercises: warm-ups (A1, A3, A5, A6) and derivations (A2, A4, A7–A11)
- ▣ **Part B:** fifteen computational exercises: verify (B1–B8), then test (B9–B15)
- ▣ **Part C:** one open question, a starting point for the team project (C1), an AI answer to critique (C2), and a replication of a landmark paper as a project seed (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every Part B exercise ends with **interpretation questions**
 - ▶ They cannot be answered by reading a number: ask what would change the conclusion
- Report every estimate with its uncertainty (standard error, interval or p-value)
- Before any test: look at the data (charts, extremes, zero returns, calendar)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (returns and annualisation), A3 (Jarque–Bera and Ljung–Box, step by step), A5 (Parkinson volatility and its efficiency), A6 (kurtosis of a mixture), A7 (the Hill estimator, step by step)
- Part B: B1 (descriptive statistics), B3 (uncorrelated but not independent), B9 (robust Ljung–Box), B11 (Hill on both tails)

Proposed: you solve them

- Part A: A2 (model: A1); A4 (model: A5); A8, A9, A10 (model: A6); A11 (model: A7)
- Part B: B2, B6, B7 (model: B1); B4, B5 (model: B3); B8 (model: A5)
- Part B, inference: B10, B12, B13 (model: B9); B14 (model: B3, B11); B15 (model: B9, B11)
- Part C: C1, open question and project seed; C2, critique an AI answer (model: A3, B3, B9); C3, replicate [Guyon & Lekeufack \(2023\)](#) (model: B3, B9)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter1_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `arch`, `matplotlib`
- Daily market data are in the course repository, folder `data/market`

```
import numpy as np, pandas as pd
URL = "https://raw.githubusercontent.com/danpele/MFM/main/data/market/"
bet = pd.read_csv(URL + "BETTR.INDX.csv", index_col="date", parse_dates=True)
r = np.log(bet["close"]).diff().dropna()           # daily log returns
```

- Other files: `BTC-USD.CC.csv`, `GSPC.INDX.csv`; EUR/RON: BNR official reference rate
- Joint analyses (correlations, Sharpe comparisons): join the *prices* on common days first, then take returns

Key Definitions: Returns and Annualisation

- ▣ **Simple return** $R_t = P_t/P_{t-1} - 1$; **log return** $r_t = \ln(1 + R_t) = \ln P_t - \ln P_{t-1}$
 - ▶ Log returns add over time; simple returns add across assets
- ▣ **Portfolio return** $R_{p,t} = \sum_i w_i R_{i,t}$, with weights w_i summing to 1
- ▣ **Annualisation** with P periods per year (252 for stocks, 365 for crypto)
 - ▶ $\mu_{\text{ann}} = P \bar{r}$ needs only a constant mean; $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$ is exact for serially uncorrelated returns with constant variance (A2)
- ▣ **CAGR** (Compound Annual Growth Rate) $= (P_T/P_0)^{1/Y} - 1$, with Y the number of years
- ▣ **h -day return**: sum of h consecutive daily log returns; non-overlapping blocks share no days

Key Definitions: Moments and Tests

- ▣ **Skewness** $\hat{S} = \frac{1}{T} \sum_t z_t^3$ and **kurtosis** $\hat{K} = \frac{1}{T} \sum_t z_t^4$, with $z_t = (r_t - \bar{r})/\hat{\sigma}$
 - ▶ Normal distribution: $S = 0$, $K = 3$; **excess kurtosis** $= K - 3$; for i.i.d. data from the Normal distribution $SE(\hat{K} - 3) \approx \sqrt{24/T}$
- ▣ **Jarque–Bera** (JB) normality test: $JB = \frac{T}{6} (\hat{S}^2 + (\hat{K} - 3)^2/4) \sim \chi_2^2$ under i.i.d. returns from the Normal distribution
- ▣ **Autocorrelation** $\hat{\rho}_k = \text{corr}(r_t, r_{t-k})$; **ACF** (AutoCorrelation Function): $\hat{\rho}_k$ against k
- ▣ **Ljung–Box** test: $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$ under i.i.d. returns
- ▣ **ARCH-LM** test (AutoRegressive Conditional Heteroskedasticity, Lagrange Multiplier): $\hat{\varepsilon}_t = r_t - \bar{r}$; nR^2 from regressing $\hat{\varepsilon}_t^2$ on a constant and q lags, $n = T - q$; asymptotically χ_q^2 under no ARCH effects (finite fourth moment)
- ▣ **p-value**: probability under H_0 of a statistic at least as extreme as observed; reject at 5% if below 0.05
- ▣ **Student-t** with ν degrees of freedom: heavy tails for small ν , the Normal distribution as $\nu \rightarrow \infty$

Key Definitions: Risk and Performance

- ▣ **VaR** (Value-at-Risk) at 1%, one day: the loss exceeded on 1% of days, $\text{VaR} = -q_{0.01}$
 - ▶ q_α : quantile, the value below which a fraction α of returns falls; a **breach**: a day whose loss is larger than the VaR
- ▣ **Drawdown** $DD_t = P_t / \max_{s \leq t} P_s - 1$; **maximum drawdown** $MDD = \min_t DD_t$, in log units
 $D = -\ln(1 + MDD)$
- ▣ **Sharpe ratio** $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$ on simple returns R_t , r_f the risk-free rate (here 0); $SE \approx 1/\sqrt{Y}$ for i.i.d. returns from the Normal distribution over Y years [Lo \(2002\)](#)
- ▣ **Sortino ratio**: as Sharpe, with the downside deviation $\sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$ (all days) in the denominator
- ▣ **Calmar ratio** $= \text{CAGR} / |MDD|$
- ▣ **Leverage correlation** $L(k) = \text{corr}(r_t, |r_{t+k}|)$: negative when losses raise future volatility

Key Definitions: Volatility Estimators and Data

- ▣ **OHLC**: Open, High, Low and Close prices of a day; o_t, h_t, l_t, c_t their logs
- ▣ **Close-to-close** volatility: standard deviation of daily log returns
- ▣ **Range-based** estimators (formulas in Lecture 1)
 - ▶ Parkinson: daily range $h_t - l_t$, assumes **zero drift**; Garman–Klass: range plus open-to-close move
 - ▶ Rogers–Satchell: **drift-independent**; Yang–Zhang: adds the overnight jump $o_t - c_{t-1}$
- ▣ **BET-TR**: Total Return version of the Bucharest Stock Exchange index BET, dividends re-invested
- ▣ **EUR/RON**: BNR reference rate; a **managed float**, i.e. a market rate with central-bank interventions

Key Definitions: Inference Tools

- ▣ **Hill estimator** [Hill \(1975\)](#) from the k largest values $X_{(1)} \geq \dots \geq X_{(k+1)}$
 - ▶ $1/\hat{\alpha} = \frac{1}{k} \sum_{i=1}^k \ln(X_{(i)}/X_{(k+1)})$, $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$; $\mathbb{E}|X|^p$ exists for $p < \alpha$
- ▣ **Robust autocorrelation variance**, $x_t = r_t - \bar{r}$: $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ (i.i.d.: $\hat{\tau}_k \approx 1/T$)
 - ▶ Robust portmanteau $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ [Lobato, Nankervis & Savin \(2001\)](#), if $\mathbb{E}r^4 < \infty$ and $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for $k \neq l$
 - ▶ $\hat{\tau}_k$ is the variance under $\rho_k = 0$; to compare non-zero ρ_1 : HAC SE of the slope of r_t on r_{t-1}
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) standard errors [Newey & West \(1987\)](#): valid under clustering
- ▣ **Block bootstrap**: resample blocks of consecutive days, keeping the dependence inside each block
- ▣ **Non-synchronous trading**: markets closing at different hours record the same news on different days [Scholes & Williams \(1977\)](#), [Dimson \(1979\)](#)
- ▣ **Memory parameter** d : $\rho_k \approx c k^{2d-1}$; local Whittle $\hat{d}$ from m Fourier frequencies, $\text{SE} = 1/(2\sqrt{m})$ [Robinson \(1995\)](#)

Exercise A1: Returns and Annualisation, a Warm-Up [Solved]

Task

- Data: hypothetical numbers, no file. A share closes at 100, 110, 99, 104.5 on four consecutive days and pays no dividend
- A portfolio holds 60% in asset A and 40% in asset B for one day; $R_A = +5\%$, $R_B = -2\%$
- An equity index has a daily mean log return of 0.04% and a daily volatility of 1.2%; Bitcoin has a daily volatility of 3.5%; returns are assumed serially uncorrelated with constant variance
- (a) Compute the three daily simple returns $R_t = P_t/P_{t-1} - 1$ and log returns $r_t = \ln(P_t/P_{t-1})$
- (b) Compute the 3-day simple and log returns from the first and the last price, and compare them with $\sum_t R_t$ and $\sum_t r_t$
- (c) Compute the portfolio simple return R_p , the weighted log return $0.6 r_A + 0.4 r_B$ and the true portfolio log return $\ln(1 + R_p)$
- (d) Annualise the index mean and volatility with $P = 252$; annualise the Bitcoin volatility and justify your choice of P

Report and interpret

- Report a table of R_t and r_t , the three portfolio numbers and the annualised values
- Interpretation: which return aggregates exactly over time, which across assets, and why must P differ between stocks and Bitcoin?

Solution A1 (1/2)

- (a) $R_1 = 110/100 - 1 = 10\%$, $R_2 = 99/110 - 1 = -10\%$, $R_3 = 104.5/99 - 1 = 5.56\%$
 - ▶ $r_1 = \ln 1.10 = 0.0953$, $r_2 = \ln 0.90 = -0.1054$, $r_3 = \ln 1.0556 = 0.0541$
- (b) 3-day simple return: $104.5/100 - 1 = 4.5\%$, but $\sum_t R_t = 10 - 10 + 5.56 = 5.56\%$: the sum of simple returns is wrong
 - ▶ 3-day log return: $\ln 1.045 = 0.0440 = 0.0953 - 0.1054 + 0.0541$: the sum of log returns is exact
- (c) $R_p = 0.6 \times 0.05 + 0.4 \times (-0.02) = 2.20\%$; weighted log return $0.6 \times 0.04879 - 0.4 \times 0.02020 = 0.02119$; true log return $\ln 1.022 = 0.02176$
- Interpretation: log returns add over time, simple returns add across assets; for daily moves the error of the wrong rule is small, for large moves it is not

Solution A1 (2/2)

- (d) Index: $\mu_{\text{ann}} = 252 \times 0.04\% = 10.08\%$; $\sigma_{\text{ann}} = \sqrt{252} \times 1.2\% = 15.87 \times 1.2\% = 19.05\%$
- Bitcoin trades every calendar day, so $P = 365$: $\sigma_{\text{ann}} = \sqrt{365} \times 3.5\% = 19.10 \times 3.5\% = 66.9\%$
 - ▶ With $P = 252$ one would report 55.6%, i.e. 17% too low
- Interpretation: P must equal the number of return observations per year; the $\sqrt{P}$ rule is exact only for serially uncorrelated returns with constant variance, and A2 shows the correction under autocorrelation

Exercise A2: The $\sqrt{h}$ Rule under Autocorrelation [Proposed]

Solution discussed in class

Task

- ▣ Assumption: daily log returns follow a stationary AR(1), $r_t = \rho r_{t-1} + e_t$ with $|\rho| < 1$, e_t white noise and $\text{Var}(r_t) = \sigma^2$, so $\gamma_j = \text{Cov}(r_t, r_{t-j}) = \sigma^2 \rho^j$
- ▣ Data for (c), from Lecture 1: EUR/RON (BNR reference rate, July 2005 – September 2026), $\hat{\rho}_1 = 0.170$, $\sqrt{252} \hat{\sigma} = 4.44\%$; S&P 500 (1990–2026), $\hat{\rho}_1 = -0.079$, $\sqrt{252} \hat{\sigma} = 17.99\%$
- ▣ (a) Write $\text{Var}(\sum_{j=1}^h r_{t+j})$ as a double sum of autocovariances and show that it equals $h\sigma^2 [1 + 2 \sum_{j=1}^{h-1} (1 - j/h)\rho^j]$
- ▣ (b) Show that the variance ratio $VR(h) = \text{Var}(\sum_{j=1}^h r_{t+j}) / (h\sigma^2)$ tends to $(1 + \rho)/(1 - \rho)$ as $h \rightarrow \infty$
- ▣ (c) Compute the exact factor $\sqrt{VR(252)}$ for both series and the corrected annual volatilities
- ▣ (d) Explain why volatility clustering alone does not change the result of (a)
- ▣ Model: A1

Exercise A2: The $\sqrt{h}$ Rule under Autocorrelation [Proposed] (continued)

Solution discussed in class

Report and interpret

- ☐ Report the two factors and the two corrected volatilities
- ☐ Interpretation: for which series does the $\sqrt{252}$ rule mislead more, and in which direction?

Exercise A3: Jarque–Bera and Ljung–Box, Step by Step [Solved]

Task

- Data: summary statistics of two hypothetical samples of daily returns; both tests are run at the 5% level
- Assumption under H_0 : i.i.d. returns; for JB they also follow the Normal distribution
- (a) With $T = 2,500$, $\hat{S} = -0.5$, $\hat{K} = 8$, compute $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ Jarque & Bera (1980) and compare it with the 5% critical value of χ_2^2 , 5.99
- (b) With $T = 1,000$ and $\hat{\rho}_1 = 0.10$, $\hat{\rho}_2 = -0.05$, $\hat{\rho}_3 = 0.03$, compute $Q(3) = T(T+2) \sum_{k=1}^3 \hat{\rho}_k^2 / (T-k)$ Ljung & Box (1978) and compare it with the critical value of χ_3^2 , 7.81
- (c) Name the stylised fact that makes $Q(3)$ too large under H_0 and the exercise that corrects it

Report and interpret

- Report each statistic, its p-value and the decision at 5%
- Interpretation: what exactly does each rejection tell you about the returns?

Solution A3

- (a) Skewness term $\hat{S}^2/6 = 0.25/6 = 0.042$; kurtosis term $(\hat{K} - 3)^2/24 = 25/24 = 1.042$
 - ▶ $JB = 2,500 \times (0.042 + 1.042) = 2,708.3$, $p < 10^{-10}$: reject normality at 5%; 96% of the statistic comes from the kurtosis term
- (b) Terms $0.01/999 = 1.001 \times 10^{-5}$, $0.0025/998 = 0.251 \times 10^{-5}$, $0.0009/997 = 0.090 \times 10^{-5}$, sum 1.342×10^{-5}
 - ▶ $Q(3) = 1,000 \times 1,002 \times 1.342 \times 10^{-5} = 13.44$, $p = 0.004$: reject “no autocorrelation” at 5%
- (c) Volatility clustering: returns are uncorrelated but not independent, so $\text{Var}(\hat{\rho}_k) > 1/T$ and Q rejects too often; B9 divides each $\hat{\rho}_k^2$ by its robust variance
- Interpretation: the JB rejection is about heavy tails, not asymmetry; the Ljung–Box rejection may be an artefact of clustering rather than predictability

Exercise A4: How Large Should a Drawdown Be? [Proposed]

Solution discussed in class

Task

- ▣ Data for (a): hypothetical year-end portfolio values 100, 120, 90, 110, 130, 117
- ▣ Assumption for (b)–(d): the log price is a driftless Brownian motion with annual volatility σ over T years; then the maximum log drawdown $D = -\ln(1 + MDD)$ has $\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ Magdon-Ismail et al. (2004)
- ▣ Data for (d): gold spot (USD per ounce, weekdays), 1990–2026: $\sigma = 16.3\%$, $T = 36.7$ years, observed $MDD = -44.7\%$ ($D = 0.592$), mean log return 6.5% a year
- ▣ (a) Compute the running peak, the drawdowns $DD_t = V_t / \max_{s \leq t} V_s - 1$, the MDD and the Calmar ratio with a 5-year CAGR of 3.19%
- ▣ (b) Use the scaling property $W(cT) \stackrel{d}{=} \sqrt{c} W(T)$ to explain why $\mathbb{E}[D]$ is proportional to $\sqrt{T}$
- ▣ (c) Compute the ratio of $\mathbb{E}[D]$ for samples of 40 and 10 years of the same process
- ▣ (d) Compute the driftless benchmark $\mathbb{E}[D]$ for gold and compare it with the observed D
- ▣ (e) State what (b)–(d) imply for Calmar ratios computed on samples of different length
- ▣ Model: A5

Exercise A4: How Large Should a Drawdown Be? [Proposed] (continued)

Solution discussed in class

Report and interpret

- ☐ Report the drawdown series, the benchmark and a one-sentence answer to (e)
- ☐ Interpretation: why is gold's drawdown smaller than the benchmark?

Exercise A5: Parkinson Volatility and Its Efficiency [Solved]

Task

- Data: the high/low ratios H/L of five days: 1.020, 1.015, 1.030, 1.010, 1.025; 252 trading days a year
- Assumption: within the day the log price is a driftless Brownian motion with volatility σ ; no overnight jumps
- (a) Compute $\ln(H/L)$ for each day, the mean of the squares, $\hat{\sigma}_P^2 = \overline{(\ln H - \ln L)^2} / (4 \ln 2)$ [Parkinson \(1980\)](#), and the daily and annual volatility
- (b) Explain how a strong drift affects $\hat{\sigma}_P$ and name the information the estimator does not use
- (c) Using $\mathbb{E}[(h - l)^2] = 4 \ln 2 \sigma^2$ and $\mathbb{E}[(h - l)^4] = 9\zeta(3) \sigma^4$ (ζ the Riemann zeta function, $\zeta(3) = 1.202$), derive $\text{Var}((h - l)^2 / (4 \ln 2))$ and compare it with $\text{Var}(r_t^2) = 2\sigma^4$ for $r_t \sim N(0, \sigma^2)$

Report and interpret

- Report the daily and annual volatility and the efficiency ratio
- Interpretation: how many close-to-close days is one day of ranges worth?

Solution A5 (1/2)

- (a) $\ln(H/L) = 0.0198, 0.0149, 0.0296, 0.0100, 0.0247$; squares ($\times 10^{-4}$) 3.92, 2.22, 8.74, 0.99, 6.10
 - ▶ Mean 0.000439; $\hat{\sigma}_P^2 = 0.000439/2.7726 = 0.000158$; $\hat{\sigma}_P = 1.26\%$ a day, $1.26\% \times 15.87 = 20.0\%$ a year
- (b) A drift widens the daily range, so $\hat{\sigma}_P$ is biased upward; [Rogers & Satchell \(1991\)](#) is drift-independent
 - ▶ The overnight jump (previous close to open) is not used; [Yang & Zhang \(2000\)](#) adds it

Solution A5 (2/2)

- (c) $\text{Var}((h - l)^2) = \mathbb{E}(h - l)^4 - [\mathbb{E}(h - l)^2]^2 = (9\zeta(3) - 16(\ln 2)^2)\sigma^4 = (10.819 - 7.687)\sigma^4 = 3.131\sigma^4$
 - ▶ Dividing by $(4 \ln 2)^2 = 7.687$: $\text{Var}(\hat{\sigma}_P^2) = 0.407\sigma^4$ per day
 - ▶ Close-to-close: $\text{Var}(r_t^2) = \mathbb{E}r_t^4 - (\mathbb{E}r_t^2)^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4$
- Efficiency $2/0.407 = 4.9$: one day of ranges carries about the information of five close-to-close days, the “about 5” of Lecture 1
- Interpretation: the gain assumes driftless continuous trading without overnight jumps; a stale open affects the estimators that use the open (Garman–Klass, Rogers–Satchell, Yang–Zhang), not Parkinson, which uses only the high and low (B8)

Exercise A6: Kurtosis of a Volatility Mixture [Solved]

Task

- Assumption: $r_t = \sigma_t \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$ i.i.d. and independent of σ_t ; $\sigma_t = 1\%$ with probability 0.8 and 2% with probability 0.2, independently across days
- (a) Using independence and $\mathbb{E}\varepsilon^4 = 3$, show that $K = \mathbb{E}r^4 / (\mathbb{E}r^2)^2 = 3 \mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2$
- (b) Compute $\mathbb{E}\sigma^2$ and $\mathbb{E}\sigma^4$ (in $\%^2$ and $\%^4$) and K
- (c) Use Jensen's inequality to show that $K \geq 3$, with equality only for a constant σ
- (d) State whether the unconditional distribution of r_t is the Normal distribution

Report and interpret

- Report K and the excess kurtosis
- Interpretation: link the result to the mixture-of-distributions hypothesis of [Clark \(1973\)](#)

Solution A6

- (a) $\mathbb{E}r^4 = \mathbb{E}\sigma^4 \mathbb{E}\varepsilon^4 = 3 \mathbb{E}\sigma^4$ and $\mathbb{E}r^2 = \mathbb{E}\sigma^2 \mathbb{E}\varepsilon^2 = \mathbb{E}\sigma^2$, hence $K = 3 \mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2$
- (b) $\mathbb{E}\sigma^2 = 0.8 \times 1 + 0.2 \times 4 = 1.6$; $\mathbb{E}\sigma^4 = 0.8 \times 1 + 0.2 \times 16 = 4.0$; $K = 3 \times 4.0 / 1.6^2 = 12 / 2.56 = 4.69$, excess kurtosis 1.69
- (c) Jensen: $\mathbb{E}[(\sigma^2)^2] \geq (\mathbb{E}\sigma^2)^2$, so $K \geq 3$; equality iff $\text{Var}(\sigma^2) = 0$
- (d) No: within each regime returns follow the Normal distribution, but the mixture has a higher peak and heavier tails
- Interpretation: random volatility alone creates heavy tails; [Clark \(1973\)](#)'s random "business time" is such a mixture

Exercise A7: The Hill Estimator, Step by Step [Solved]

Task

- Data: the 11 largest absolute daily returns (%) of a hypothetical sample: 12.0; 9.5; 8.2; 7.4; 6.9; 6.1; 5.8; 5.5; 5.2; 5.0; 4.8
- Assumption: above the threshold $X_{(k+1)}$ the tail is Pareto, $P(X > x) \propto x^{-\alpha}$, and the observations are independent
- (a) For $k = 10$ compute the ten log-ratios $\ln(X_{(i)}/X_{(11)})$, their mean and $\hat{\alpha} = 1/\text{mean}$ [Hill \(1975\)](#)
- (b) Compute $\text{SE} = \hat{\alpha}/\sqrt{k}$ and the 95% interval $\hat{\alpha} \pm 1.96 \text{ SE}$
- (c) State which moments exist at the point estimate and which claims hold with 95% confidence
- (d) Explain why choosing a very large k is not a remedy
- (e) Repeat (a)–(b) on the 11 largest $|r_t|$ of BET-TR (BETTR.INDX, 2014–2026)

Report and interpret

- Report $\hat{\alpha}$, its SE and interval for both samples
- Interpretation: can you decide whether the fourth moment exists?

Solution A7 (1/2)

- (a) Log-ratios to 4.8: 0.916; 0.683; 0.536; 0.433; 0.363; 0.240; 0.189; 0.136; 0.080; 0.041
 - ▶ Sum 3.616, mean 0.3616, so $\hat{\alpha} = 1/0.3616 = 2.77$
- (b) $SE = 2.77/\sqrt{10} = 0.87$; interval $2.77 \pm 1.96 \times 0.87 = [1.05; 4.48]$
- (c) At the point estimate the variance exists ($2.77 > 2$) and the fourth moment does not ($2.77 < 4$)
 - ▶ But the interval contains values below 2 and above 4: with $k = 10$ neither claim holds at 95%

Solution A7 (2/2)

- (d) A large k uses observations from the body, where the power law fails: less variance, more bias; read the Hill plot where it is flat
- (e) BET-TR: the largest day is -11.86% on 19 December 2018, the 11th largest 5.20% ; log-ratios sum to 2.958
 - ▶ $\hat{\alpha}_{10} = 10/2.958 = 3.38$, $SE = 3.38/\sqrt{10} = 1.07$, interval $[1.28; 5.48]$
- Interpretation: ten observations cannot decide whether the fourth moment exists; B11 uses $k = 2\%$ of n

Exercise A8: Where the Autocorrelation of r_t^2 Comes From [Proposed]

Solution discussed in class

Task

- Assumption: as in A6, $r_t = \sigma_t \varepsilon_t$ with $\sigma_t \in \{1\%, 2\%\}$ and stationary probabilities (0.8; 0.2), but the regime is a two-state Markov chain [Hamilton \(1989\)](#) with $P(\text{high} \rightarrow \text{high}) = p_{HH} = 0.9$; ε_t i.i.d. $N(0, 1)$, independent of the chain
- Data for (d): S&P 500, 1990–2026, ACF of r_t^2 about 0.31 at lag 1 and 0.04 at lag 100 (Lecture 1)
- (a) Use the balance condition $\pi_L p_{LH} = \pi_H p_{HL}$ to find p_{LH} and p_{LL}
- (b) Show that $\text{Cov}(r_t^2, r_{t-k}^2) = \text{Var}(\sigma_t^2) \lambda^k$ for $k \geq 1$, with $\lambda = p_{LL} + p_{HH} - 1$
- (c) Compute $\text{Var}(\sigma^2)$, $\text{Var}(r^2)$ and $\text{corr}(r_t^2, r_{t-k}^2)$ at lags 1 and 10
- (d) Compare with the S&P 500 and name two features the model cannot reproduce
- Model: A6

Report and interpret

- Report the transition probabilities and the two autocorrelations
- Interpretation: what kind of volatility model would close the gap?

Exercise A9: Can a Gaussian GARCH Produce the S&P 500 Tails? [Proposed]

Solution discussed in class

Task

- ▣ Model: GARCH(1,1) with $N(0, 1)$ innovations, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ [Bollerslev \(1986\)](#); full treatment in Chapter 5
- ▣ Result used: if $1 - (\alpha + \beta)^2 - 2\alpha^2 > 0$, the unconditional kurtosis is $K = 3[1 - (\alpha + \beta)^2]/[1 - (\alpha + \beta)^2 - 2\alpha^2]$
- ▣ Data for (d): S&P 500 daily log returns 1990–2026 (GSPC.INDX), excess kurtosis 10.9, i.e. $K = 13.9$
- ▣ (a) Compute K for $(\alpha, \beta) = (0.10; 0.85)$ and $(0.10; 0.88)$
- ▣ (b) Derive from the formula the condition for a finite fourth moment
- ▣ (c) With $\alpha + \beta = 0.99$ fixed, solve for the α that gives $K = 13.9$ and compute the denominator
- ▣ (d) Fit a Gaussian GARCH(1,1) to the S&P 500 (package arch, returns in %) and compute its implied K
- ▣ Model: A6

Exercise A9: Can a Gaussian GARCH Produce the S&P 500 Tails? [Proposed] (continued)

Solution discussed in class

Report and interpret

- ☐ Report the four kurtosis values and the parameters of (c) and (d)
- ☐ Interpretation: what do you conclude about the distribution of the standardised residuals?

Exercise A10: Why the Autocorrelation Band Widens under Clustering [Proposed]

Solution discussed in class

Task

- Assumption: $x_t = \sigma_t \varepsilon_t$, ε_t i.i.d. $N(0, 1)$, independent of the whole process $\{\sigma_s\}$, which is stationary and ergodic with bounded σ_t (as in A6 and A8); then x_t is a martingale difference and $T \text{Var}(\hat{\rho}_1) \rightarrow T\tau_1 = \mathbb{E}[x_t^2 x_{t-1}^2] / (\mathbb{E}x_t^2)^2$ [Lobato, Nankervis & Savin \(2001\)](#)
- Data for (c): S&P 500, 1990–2026, $T = 9,245$, $\hat{\rho}_1 = -0.079$, $T\hat{\tau}_1 = 5.00$ (Lecture 1)
- (a) For the i.i.d. mixture of A6 (σ_t independent across days) show that $T\tau_1 = 1$
- (b) For the Markov regimes of A8 show that $\mathbb{E}[x_t^2 x_{t-1}^2] = (\mathbb{E}\sigma^2)^2 + \lambda \text{Var}(\sigma^2)$; compute $T\tau_1$ and the band-widening factor $\sqrt{T\tau_1}$
- (c) Compute the i.i.d. and the robust t -statistics of the S&P 500 lag-1 autocorrelation
- Model: A6

Report and interpret

- Report $T\tau_1$ for the two models and the two t -statistics
- Interpretation: which feature of returns widens the band, heavy tails or clustering?

Exercise A11: The Hill Estimator as a Maximum Likelihood Estimator [Proposed]

Solution discussed in class

Task

- Assumption: given the threshold $u = X_{(k+1)}$, the k ordered exceedances $X_{(1)}, \dots, X_{(k)}$ are the order statistics of an i.i.d. Pareto sample, $f(x) = \alpha u^\alpha x^{-\alpha-1}$ for $x > u$
- (a) Write the log-likelihood $\ell(\alpha)$ and show that its maximiser is the Hill estimator $\hat{\alpha} = k / \sum_{i=1}^k \ln(X_{(i)}/u)$ Hill (1975)
- (b) Compute the Fisher information $-\mathbb{E} \ell''(\alpha)$ and derive $\text{SE}(\hat{\alpha}) \approx \hat{\alpha} / \sqrt{k}$
- (c) Check (a)–(b) on the data of A7
- (d) Simulate 2,000,000 absolute values from a Student- t with $\nu = 3$ (tail index 3; seed 1) and compute $\hat{\alpha}$ for $k = 200, 2,000, 20,000$ and $200,000$
- Model: A7

Report and interpret

- Report the derivation, $\hat{\alpha}$ and SE for A7 and the four simulated estimates
- Interpretation: why does $\hat{\alpha}$ drift away from 3 when k is large?

Exercise B1: Descriptive Statistics [Solved]

Task

- Data: daily closes of BET-TR (BETTR.INDX, 24 Sep 2014 – 18 Sep 2026, 2,992 returns), Bitcoin (BTC-USD.CC, 18 Sep 2014 – 18 Sep 2026, 4,384 returns) and EUR/RON (BNR reference rate, 5 Jul 2005 – 18 Sep 2026, 5,352 returns)
- Daily log returns on each series' own calendar; $P = 252$ (BET-TR, EUR/RON) and 365 (Bitcoin)
- (a) Compute the mean, standard deviation, annualised volatility, skewness, excess kurtosis, minimum and maximum
- (b) Run the Jarque–Bera test for each series at the 5% level
- (c) Recompute the BET-TR excess kurtosis after removing the day with the largest absolute return and after removing the five days with the largest absolute returns
- (d) Compute the Hinkley skewness and the Crow–Siddiqui kurtosis [Kim & White \(2004\)](#), with 95% intervals from a moving-block bootstrap (blocks of 20 days, 1,000 resamples)

Report and interpret

- Report one table for (a)–(b) and one for (d)
- Interpretation: which conclusion survives (c) and (d): heavy tails, negative skewness, or both?

Solution B1 (1/2)

	BET-TR	Bitcoin	EUR/RON
Mean / std. dev. (% a day)	0.077 / 0.978	0.118 / 3.485	0.007 / 0.279
Annualised volatility (%)	15.5	66.6	4.4
Skewness	−1.40	−0.70	0.73
Excess kurtosis	17.8	12.0	15.1
Min (% , date)	−11.9 (19 Dec 2018)	−46.5 (12 Mar 2020)	−2.5 (9 Oct 2008)
Max (%)	6.7	22.5	2.9
JB	40,433	26,482	51,575

- (a) Annualised volatility = $\sqrt{P} \hat{\sigma}$: e.g. BET-TR $\sqrt{252} \times 0.978\% = 15.5\%$, Bitcoin $\sqrt{365} \times 3.485\% = 66.6\%$
- (b) All JB statistics exceed 5.99 by four orders of magnitude, $p < 10^{-10}$: normality is rejected for the three series
- (c) BET-TR excess kurtosis: 17.8 with all days, 11.8 without 19 December 2018 (largest $|r_t|$), 5.95 without the five largest $|r_t|$: one day carries a third of the statistic

Solution B1 (2/2)

- (d) Hinkley = $\frac{q0.95 + q0.05 - 2q0.5}{q0.95 - q0.05}$, Crow–Siddiqui = $\frac{q0.975 - q0.025}{q0.75 - q0.25} - 2.91$ (both 0 for the Normal distribution)

	BET-TR	Bitcoin	EUR/RON
Hinkley skewness [95% CI]	−0.036 [−0.085, 0.025]	−0.018 [−0.069, 0.032]	0.041 [−0.022, 0.093]
Crow–Siddiqui kurtosis [95% CI]	1.06 [0.73, 1.43]	2.25 [1.88, 2.64]	4.87 [3.74, 6.13]

- Heavy tails survive: every kurtosis interval lies above 0 and the fragile moment kurtosis is replaced by a stable quantile measure
- Skewness does not survive: every interval contains 0; BET-TR's −1.40 comes from a few crash days
- Interpretation: report heavy tails with robust measures and tail indices; JB cannot separate the two conclusions because it mixes both terms

Exercise B2: Heavy Tails and VaR 1% on BET-TR [Proposed]

Solution discussed in class

Task

- Data: BET-TR daily log returns (`BETTR.INDX`), 24 Sep 2014 – 18 Sep 2026, $T = 2,992$; level $\alpha = 1\%$, horizon one day
- Design: the models are estimated and checked on the **same** sample: a descriptive, in-sample comparison, not a backtest
- (a) Standardise $z_t = (r_t - \bar{r})/\hat{\sigma}$ and count the share of days with $|z_t| > 3$; compare it with 0.27% under the Normal distribution
- (b) Fit a Student- t (degrees of freedom ν , location, scale) to z_t by maximum likelihood (`scipy.stats.t.fit`)
- (c) Compute $\text{VaR}_{1\%} = -q_{0.01}$ in three ways: Normal distribution $\bar{r} + \hat{\sigma} z_{0.01}$; Student- t $\bar{r} + \hat{\sigma} q_{0.01}^t$; the empirical 1% quantile
- (d) Count the breaches (days with $r_t < q_{0.01}$) of each VaR and compare them with $0.01 T \approx 30$
- Model: B1

Exercise B2: Heavy Tails and VaR 1% on BET-TR [Proposed] (continued)

Solution discussed in class

Report and interpret

- Report a table: VaR and number of breaches for the three methods
- Interpretation: why does the empirical VaR have about 30 breaches by construction, and how would you turn this into a genuine backtest (Chapter 8)?

Exercise B3: Uncorrelated but not Independent (Bitcoin) [Solved]

Task

- ▣ Data: Bitcoin daily log returns (BTC-USD.CC), 18 Sep 2014 – 18 Sep 2026, $T = 4,384$; tests at 5%
- ▣ (a) Plot the ACF of r_t and of $|r_t|$ for lags 1–100 with the i.i.d. band $\pm 1.96/\sqrt{T} = \pm 0.030$
- ▣ (b) Compute the Ljung–Box $Q(10)$ on r_t and on r_t^2 , with p-values
- ▣ (c) Run the ARCH-LM test with $q = 5$ lags [Engle \(1982\)](#): with $\hat{\varepsilon}_t = r_t - \bar{r}$, regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$ and use $(T - 5)R^2$, asymptotically χ_5^2
- ▣ (d) Compare the p-value of $Q(10)$ on r_t with that of the robust $\tilde{Q}(10)$ of B9

Report and interpret

- ▣ Report the lag-1 autocorrelations, the three statistics with p-values and the robust p-value
- ▣ Interpretation: the p-value of $Q(10)$ on r_t is about 3%: is that evidence of predictability, given what $Q(10)$ on r_t^2 says about the i.i.d. assumption?

Solution B3

- (a) $\hat{\rho}_1(r_t) = -0.022$, inside the band; $\hat{\rho}_1(|r_t|) = 0.213$, seven times the band, and positive for many lags
- (b) $Q(10)$ on r_t : 20.2, $p = 0.028$: rejected at 5% under i.i.d.; on r_t^2 : 212.7, $p \approx 4 \times 10^{-40}$
- (c) $\text{ARCH-LM}(5) = 114.7$, $p \approx 4 \times 10^{-23}$: strong conditional heteroskedasticity (the χ^2 reference needs a finite fourth moment, so the p-value is indicative)
- (d) The robust $\tilde{Q}(10)$ has $p = 0.29$: no evidence of linear autocorrelation at lags 1–10
- Interpretation: (b) and (c) show that returns are not i.i.d., so the 3% p-value of the classical $Q(10)$ is not valid; these tests find no evidence of linear return predictability at the tested lags, while return magnitudes show dependence

 MFM_ch1_volatility_clustering

Exercise B4: Aggregational Gaussianity (Bitcoin) [Proposed]

Solution discussed in class

Task

- Data: Bitcoin daily log returns, 2014–2026, $T = 4,384$
- (a) Build non-overlapping h -day log returns (sums of h consecutive daily returns, starting on the first day) for $h = 1, 5, 20$; drop the incomplete last block
- (b) Compute the excess kurtosis of each series and the approximate standard error $\sqrt{24/N}$, N the number of blocks
- (c) Explain why overlapping windows must not be used here
- Model: B3

Report and interpret

- Report a table with h , N , excess kurtosis and $\sqrt{24/N}$
- Interpretation: $h = 60$ is a quarter for stocks but two months for Bitcoin: how would you compare horizons across markets? Why is $\sqrt{24/N}$ too optimistic here?

Exercise B5: Leverage Effect — Romania vs Crypto [Proposed]

Solution discussed in class

Task

- Data: daily log returns of BET-TR and Bitcoin, 2014–2026, each on its own calendar
- (a) Compute the leverage correlation $L(k) = \text{corr}(r_t, |r_{t+k}|)$ for $k = -20, \dots, 20$ [Christie \(1982\)](#) and plot both series with the i.i.d. reference band $\pm 1.96/\sqrt{T}$ (pointwise; not a valid confidence band under volatility clustering)
- (b) Report $L(1)$ and $L(5)$ for both markets and check them against the band (0.036 for BET-TR, 0.030 for Bitcoin)
- Model: B3

Report and interpret

- Report the chart and the four values
- Interpretation: why is the leverage effect weaker for Bitcoin? Is a correlation enough, or do you need a regression with controls (B12)?

Exercise B6: EUR/RON Across Regimes [Proposed]

Solution discussed in class

Task

- Data: EUR/RON BNR reference rate, 5 Jul 2005 – 18 Sep 2026, daily log returns, split into July 2005–2011, 2012–2019 and 2020–2026
- (a) For each period compute $\hat{\rho}_1$ as the slope of r_t on r_{t-1} , its HAC SE Newey & West (1987) (lag truncation $\lfloor 4(T/100)^{2/9} \rfloor$), the annualised volatility ($P = 252$) and the excess kurtosis; $\sqrt{\hat{\tau}_1}$ is valid only under $\rho_1 = 0$
- (b) Locate the single variance break with the cumulative sum of squares Inclán & Tiao (1994): $D_k = C_k/C_T - k/T$, $C_k = \sum_{t \leq k} x_t^2$, break at $\arg \max_k |D_k|$
- (c) Test H_0 : ρ_1 equal in the three periods with $W = \sum_j (\hat{\rho}_j - \bar{\rho})^2 / \text{SE}_j^2$, $\bar{\rho}$ the precision-weighted mean, $W \sim \chi_2^2$, at 5%
- Model: B1

Report and interpret

- Report the table of (a), the break date and W with its p-value
- Interpretation: why is the 2020–2026 SE so large? Which statistic would change if you used market quotes instead of the fixing?

Exercise B7: BET-TR vs S&P 500, Same Period [Proposed]

Solution discussed in class

Task

- Data: BET-TR (BETTR.INDX, total return, RON) and S&P 500 (GSPC.INDX, price index, USD); join the *prices* on common dates (23 Sep 2014 – 18 Sep 2026, 2,903 returns), then compute returns; $r_f = 0$, $P = 252$
- (a) Compute the CAGR, the annualised volatility of simple returns, the Sharpe ratio $252\bar{R}/(\sqrt{252}\hat{\sigma}_R)$, the Sortino ratio (downside deviation over all days) [Sortino & van der Meer \(1991\)](#), the maximum drawdown and the Calmar ratio
- (b) Compute the correlation of daily log returns on the same day and with the S&P 500 lagged one day
- Model: B1

Report and interpret

- Report a two-column table and the two correlations
- Interpretation: is the comparison fair (dividends, currency, risk-free rate in RON)? Why is the same-day correlation too low?

Exercise B8: Range-Based Estimators on the S&P 500 [Proposed]

Solution discussed in class

Task

- Data: S&P 500 index open, high, low, close (GSPC.INDX), 2 Jan 2008 – 18 Sep 2026 (index opens before 2007 are stale); SPY ETF (SPY.US) for comparison, with open, high and low multiplied by `adjusted_close/close` so that all four prices are on the same adjusted basis
- (a) Compute 21-day rolling close-to-close, Parkinson [Parkinson \(1980\)](#), Garman–Klass [Garman & Klass \(1980\)](#), Rogers–Satchell [Rogers & Satchell \(1991\)](#) and Yang–Zhang [Yang & Zhang \(2000\)](#) volatilities, annualised with $P = 252$, and their time averages
- (b) For the index and for SPY compute the averages of the 21-day close-to-close and Yang–Zhang estimates and the full-sample overnight volatility $\text{sd}(o_t - c_{t-1})\sqrt{252}$
- Model: A5

Report and interpret

- Report the five averages and the index vs SPY comparison
- Interpretation: Yang–Zhang includes the overnight jump, yet stays below close-to-close on the index: which property of the index open explains this (Lecture 1)?

Exercise B9: Heteroskedasticity-Robust Ljung–Box [Solved]

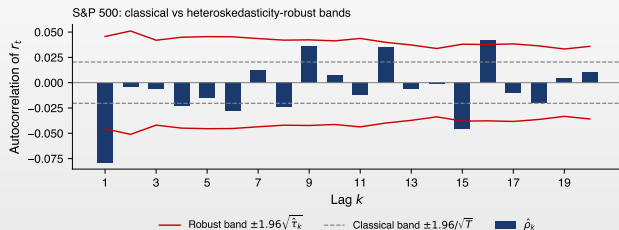
Task

- Data: daily log returns of the S&P 500 (1990–2026, $T = 9,245$), BET-TR (2014–2026, $T = 2,992$) and EUR/RON (2005–2026, $T = 5,352$); $x_t = r_t - \bar{r}$
- (a) Compute $\hat{\rho}_k$, $k = 1, \dots, 20$, with the classical band $\pm 1.96/\sqrt{T}$ and the robust band $\pm 1.96\sqrt{\hat{\tau}_k}$, $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ Lobato, Nankervis & Savin (2001)
- (b) Compute the classical $Q(10)$ and the robust $\tilde{Q}(10) = \sum_{k=1}^{10} \hat{\rho}_k^2 / \hat{\tau}_k$; compare both with χ_{10}^2 (5% critical value 18.3)
- (c) Count the lags outside each band

Report and interpret

- Report $Q(10)$, $\tilde{Q}(10)$, the p-values and the significant lags for the three series
- Interpretation: for which series does autocorrelation survive the correction? What does this say about the Ljung–Box column of Lecture 1? $\hat{\tau}_k$ also needs a finite fourth moment: what do the Hill estimates say?

Solution B9 (1/2)



- (a) S&P 500, lag 1: $\hat{\rho}_1 = -0.079$; $T\hat{\tau}_1 = 5.00$, so the robust band is $1.96\sqrt{5.00/9245} = 0.046$ instead of $1.96/\sqrt{9245} = 0.020$
- The t -statistic falls from $-0.079 \times \sqrt{9245} = -7.6$ to $-0.079/\sqrt{5.00/9245} = -3.4$: still significant, but half as strong

Solution B9 (2/2)

	S&P 500	BET-TR	EUR/RON
$Q(10)$ (p)	91.6 ($< 10^{-10}$)	70.0 ($< 10^{-10}$)	221.7 ($< 10^{-10}$)
$\tilde{Q}(10)$ (p)	19.0 (0.041)	16.9 (0.076)	37.4 (< 0.001)
Lags outside, classical / robust	8 / 3	5 / 2	6 / 2
Robust lags	1, 15, 16	2, 10	1, 3

- (b) At 5%: the classical test rejects for all three; the robust test rejects for the S&P 500 (barely) and EUR/RON, not for BET-TR
- Interpretation: only EUR/RON keeps strong autocorrelation ($\hat{\rho}_1 = 0.17$), from the managed fixing, not from market inefficiency; the Ljung–Box column of Lecture 1 overstates significance
- $\tilde{Q}$ needs $\mathbb{E}r^4 < \infty$ and zero cross terms $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}]$; the thresholded test of [Dalla, Giraitis & Phillips \(2022\)](#) finds no significant cross terms for the S&P 500 and BET-TR, but with Hill $\hat{\alpha} \approx 3$ the fourth moment is doubtful: the robust p-values are conditional on it

Exercise B10: A Kurtosis That Does Not Converge [Proposed]

Solution discussed in class

Task

- Data: S&P 500 daily log returns, 2 Jan 1990 – 18 Sep 2026 ($T = 9,245$); expanding windows = the first n years, $n = 1, \dots, 36$
- Simulation: 500 Student- t series of length T for $\nu = 3$ and for $\nu = 6$ (seed 42)
- (a) Compute the excess kurtosis of the S&P 500 on each expanding window
- (b) For each ν , compute the same curve on every simulated path and its 5%, 50% and 95% quantiles at each n
- (c) Plot the real curve over both bands (symmetric log scale)
- Model: B9

Report and interpret

- Report the chart, the three largest jumps of the real curve and the two bands at $n = 36$
- Interpretation: which days cause the jumps? Why does the curve settle for $\nu = 6$ and not for $\nu = 3$? Which band contains the S&P 500, and what does “excess kurtosis = 10.9” mean then? Which tail statistic would you report instead?

Exercise B11: Hill on Both Tails, BET-TR and Bitcoin [Solved]

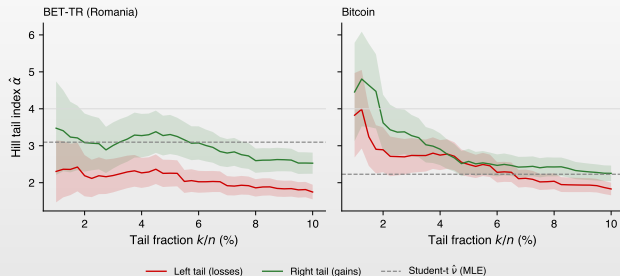
Task

- Data: daily log returns of BET-TR ($n = 2,992$) and Bitcoin ($n = 4,384$), 2014–2026; losses $-r_t$ and gains r_t treated separately; the standard errors assume independent tail observations
- (a) Plot the Hill estimate $\hat{\alpha}(k)$ for each tail, k from 1% to 10% of n , with the band $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$
- (b) Compute $\hat{\alpha}$ at $k = 2\%$ of n (BET-TR $k = 59$, Bitcoin $k = 87$) with 95% intervals
- (c) Test $H_0: \alpha_- = \alpha_+$ with $z = (\hat{\alpha}_- - \hat{\alpha}_+)/\sqrt{\hat{\alpha}_-^2/k + \hat{\alpha}_+^2/k}$, two-sided, 5%
- (d) Compare with the Student- t $\hat{\nu}$ of B2 (3.10 for BET-TR, 2.23 for Bitcoin)

Report and interpret

- Report the Hill plots, the four estimates, z and p
- Interpretation: does the test confirm the gain/loss asymmetry? With clustered extremes, is the true SE larger or smaller? Which of $\hat{\alpha}$ and $\hat{\nu}$ describes the tail?

Solution B11 (1/2)



□ (b) BET-TR: $\hat{\alpha}_- = 2.19$ [1.63; 2.75], $\hat{\alpha}_+ = 3.08$ [2.30; 3.87]; Bitcoin: $\hat{\alpha}_- = 2.89$ [2.28; 3.50], $\hat{\alpha}_+ = 3.62$ [2.86; 4.38]

► E.g. BET-TR losses: $2.19 \times (1 \pm 1.96/\sqrt{59}) = 2.19 \times (1 \pm 0.255)$

Solution B11 (2/2)

- (c) BET-TR: $\sqrt{2.19^2/59 + 3.08^2/59} = \sqrt{0.0812 + 0.1612} = 0.492$; $z = (2.19 - 3.08)/0.492 = -1.82$, $p = 0.069$
 - ▶ Bitcoin: $\sqrt{2.89^2/87 + 3.62^2/87} = 0.497$; $z = (2.89 - 3.62)/0.497 = -1.47$, $p = 0.14$
- Decision: losses are heavier in both markets, but H_0 is not rejected at 5%; clustered extremes make the true SE larger, so the evidence is weaker still
- (d) $\hat{\nu}$ is a whole-distribution estimate, valid if the Student- t fits centre and tails together; Bitcoin's $\hat{\nu} = 2.23$ lies below both tail indices, a sign that it does not; $\hat{\alpha}$ uses only the tail

Exercise B12: The Leverage Effect as a Regression [Proposed]

Solution discussed in class

Task

- ▣ Data: daily log returns of the S&P 500, BET-TR, Bitcoin and EUR/RON (full samples)
- ▣ Model: $|r_{t+1}| = c + \beta^+ r_t^+ + \beta^- r_t^- + \sum_{j=1}^5 \delta_j |r_{t-j}| + u_{t+1}$, $r_t^+ = \max(r_t, 0)$, $r_t^- = \min(r_t, 0)$; symmetric response: $\beta^+ = -\beta^-$
- ▣ (a) Estimate by OLS with Newey–West standard errors [Newey & West \(1987\)](#), lag truncation $\lfloor 4(T/100)^{2/9} \rfloor$ (8–10 here)
- ▣ (b) Test $H_0: \beta^+ + \beta^- = 0$ with a Wald χ_1^2 test at 5%
- ▣ (c) Explain why $|r_t|$ is not among the controls
- ▣ Model: B9

Report and interpret

- ▣ Report $\hat{\beta}^+$, $\hat{\beta}^-$ with SEs, the Wald statistic and p for the four markets
- ▣ Interpretation: which sign do you expect for EUR/RON, and why? Why control for $|r_{t-j}|$? Which Chapter 5 model has this asymmetry?

Exercise B13: Is the Sharpe Difference Real? (extends B7) [Proposed]

Solution discussed in class

Task

- Data: as in B7, BET-TR and S&P 500 on 2,903 common days (23 Sep 2014 – 18 Sep 2026); Sharpe ratios on simple returns, $r_f = 0$: 1.36 vs 0.75
- (a) Compute the Jobson–Korkie test [Jobson & Korkie \(1981\)](#) with the correction of [Mommel \(2003\)](#) on daily Sharpe ratios:
$$z = (\widehat{SR}_1 - \widehat{SR}_2) / \sqrt{\widehat{V}}, \quad \widehat{V} = \frac{1}{T} \left[2(1 - \hat{\rho}) + \frac{1}{2}(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\hat{\rho}^2) \right],$$
 two-sided at 5%
- (b) Compute a 95% percentile interval for the annualised difference with a circular block bootstrap (blocks of 21 days, 2,000 resamples); [Ledoit & Wolf \(2008\)](#) recommend a studentised version
- (c) Compute the correlation of daily log returns on the same day, with the S&P 500 of the previous and of the next day, and the correlation of weekly (Friday) returns
- Model: B9

Exercise B13: Is the Sharpe Difference Real? (extends B7) [Proposed] (continued)

Solution discussed in class

Report and interpret

- Report z , p , the bootstrap interval and the five correlations
- Interpretation: is the difference significant? What do the Memmel test and the bootstrap assume, and is it plausible here? Why is the weekly correlation close to the sum of the three daily ones? Which correlation would you use in a portfolio? What else makes the comparison unfair?

Exercise B14: Which Stylised Facts Does GARCH- t Reproduce? [Proposed]

Solution discussed in class

Task

- ▣ Data: S&P 500 daily log returns stored as decimals (r , as in the setup) and converted to % in the code, 1990–2026 ($T = 9,245$)
- ▣ (a) Fit a GARCH(1,1) with Student- t innovations (package `arch`, constant mean)
- ▣ (b) Simulate 200 paths of length T from the fitted model (1,000 burn-in days, fixed seed)
- ▣ (c) For each path compute the excess kurtosis, the ACF of $|r_t|$ at lags 1, 20, 100 and 250, the Hill $\hat{\alpha}$ of $|r_t|$ at $k = 2\%$ of T and the leverage correlation $L(1)$
- ▣ (d) Build 5–95% bands and mark each real value “reproduced” if it lies inside

```
from arch import arch_model
am = arch_model(100 * r, vol='GARCH', p=1, q=1, dist='t')
res = am.fit(dispatch='off')
sim = am.simulate(res.params, nobs=len(r), burn=1000)
```

- ▣ Model: B3, B11

Exercise B14: Which Stylised Facts Does GARCH- t Reproduce? [Proposed] (continued)

Solution discussed in class

Report and interpret

- Report the fitted parameters and the “reproduced / not reproduced” table
- Interpretation: does the ACF of $|r_t|$ survive at long lags, although a stationary GARCH has geometrically decaying memory? Can a symmetric GARCH produce the leverage effect? Which component would you change for each fact not reproduced?

Exercise B15: Long Memory or Breaks? [Proposed]

Solution discussed in class

Task

- Data: $|r_t|$ of the S&P 500 (1990–2026), BET-TR and Bitcoin (2014–2026), daily
- (a) Estimate the memory parameter d by local Whittle Robinson (1995) with $m = \lfloor T^{0.65} \rfloor$ Fourier frequencies; give the 95% interval with $SE = 1/(2\sqrt{m})$
- (b) Locate one variance break of r_t with the cumulative sum of squares Inclán & Tiao (1994); re-estimate $\hat{d}$ on each side and compute $z = (\hat{d}_{\text{after}} - \hat{d}_{\text{before}}) / \sqrt{SE_1^2 + SE_2^2}$, a descriptive measure: the break date is estimated from the same data
- (c) Repeat (a) with $m = T^{0.5}$ and $m = T^{0.8}$
- Model: B9, B11

Report and interpret

- Report one table with $\hat{d}$, interval, break date, $\hat{d}$ before/after, z and the sensitivity to m
- Interpretation: is $\hat{d} > 0$ robust to the break and to m ? If $\hat{d}$ changes after the break, is it long memory or a sequence of regimes Diebold & Inoue (2001)? Why does a larger m lower the SE but risk bias?

C1: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? [Proposed]

Solution discussed in class

Task

- ▣ Context: BET-TR has a positive lag-1 autocorrelation, typical of less liquid markets; Romania became a FTSE Russell Secondary Emerging market on 21 September 2020
- ▣ Data: BET-TR daily log returns (BETTR.INDX), 24 Sep 2014 – 18 Sep 2026; liquidity proxy: daily traded value (volume $\times$ close) of 11 blue chips in data/market (TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL)
- ▣ (a) Compute $\hat{\rho}_1$ on rolling one-year windows (250 days, step 5 days) with the robust bands of B9
- ▣ (b) Compare $\hat{\rho}_1$ before and after the reclassification with HAC SEs (as in B6); add a regression $r_t = a + b r_{t-1} + c \text{post}_t + e r_{t-1} \text{post}_t$ with HAC errors (10 lags)
- ▣ (c) Relate the rolling $\hat{\rho}_1$ to the log of the 250-day average traded value
- ▣ Deliverable: a Quantlet folder in the team repository and one slide with the conclusion and its main limitation
- ▣ Model: B9

C1: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? [Proposed] (continued)

Solution discussed in class

Report and interpret

- ☐ Report the chart, the before/after test, the interaction coefficient and the liquidity correlation
- ☐ Interpretation: did autocorrelation fall? Is liquidity the explanation? Discuss alternatives (index composition, new listings such as Hidroelectrica in 2023) and the link with the Adaptive Markets Hypothesis (Chapter 2)

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Are the daily BET-TR log returns Normally distributed? Are they autocorrelated? Give Python code and a conclusion”
- **The answer:** the code below, followed by the conclusion

```
from scipy import stats
from statsmodels.stats.diagnostic import acorr_ljungbox
r = np.log(bet["close"]).diff().dropna()      # daily log returns
k = stats.kurtosis(r)                          # sample kurtosis
print("excess_kurtosis:", k - 3)               # the Normal has kurtosis 3
jb, p = stats.jarque_bera(r)
lb = acorr_ljungbox(r**2, lags=[10])          # Ljung-Box, 10 lags
```

- **Conclusion of the AI**
 - ▶ “Excess kurtosis 14.8: heavy tails”
 - ▶ “JB p -value 0.000: the probability that BET-TR returns follow the Normal distribution is 0%”
 - ▶ “Ljung-Box $Q(10) = 713.5$, $p < 0.001$: returns are strongly autocorrelated, so BET-TR is predictable and not a random walk”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- (a) Find the errors: there are two or three; the rest of the answer is correct
- (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation
- (c) Give the corrected code and the corrected conclusion
- (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Model: A3, B3 and B9

Why this exercise

- AI code usually runs without an error message: the mistakes are in the statistics, not in the syntax
- Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

C3: Is the VIX Mostly Path-Dependent? [Proposed]

Solution discussed in class

Task

- ▣ Context: Guyon & Lekeufack (2023) show that an affine function of a kernel-weighted sum of past S&P 500 returns and of the square root of a kernel-weighted sum of their squares explains 94.6% of the variance of the VIX in 2000–2018 and 85.5% in 2019 – May 2022 (Table 2); they push further the finding of Bollerslev, Hood, Huss & Pedersen (2018) that past volatility at several horizons predicts future volatility (Lecture 1, Case Study)
- ▣ Data: daily closes of the S&P 500 (GSPC.INDX), the VIX (VIX.INDX) and the 9-day VIX (VIX9D.INDX) from data/market; use simple returns $r_t = S_t/S_{t-1} - 1$ on the S&P 500 calendar and divide the volatility indices by 100
- ▣ (a) For each day t , collect the last 1,000 returns (from 1996 for the first training days) and write $R_{1,t}$ and Σ_t as functions of the parameters of the TSPL kernels of eq. (3.9) of the paper, normalised on the 1,000-day window, with $\Delta t = 1/252$
- ▣ (b) Estimate the seven parameters of eq. (3.7) by nonlinear least squares on 1 Jan 2000 – 31 Dec 2018, recomputing $R_{1,t}$ and Σ_t at every trial value of (α_j, δ_j) , starting from the VIX row of Table 3, and compare your estimates with that row
- ▣ (c) Compute r^2 and RMSE on the training set and on 1 Jan 2019 – 15 May 2022 for the full model and for the two restricted models with $\beta_1 = 0$ and with $\beta_2 = 0$, each re-estimated on the training set; repeat the full model for the 9-day VIX, with a training set that starts in 2011; compare with the VIX and 9-day VIX rows of Table 2
- ▣ (d) For the VIX keep the parameters of (b) fixed, for the 9-day VIX those estimated in (c); compute r^2 and RMSE from 16 May 2022 to the last available date
- ▣ Deliverable: a Quantlet folder in the team repository and one slide with your Table 2 next to the paper's
- ▣ Model: B3 and B9

C3: Is the VIX Mostly Path-Dependent? [Proposed] (continued)

Solution discussed in class

Report and interpret

- Report the parameter table, the scores of (c) and (d), the observed and fitted VIX on the test set and the ratio of observed to fitted VIX over time
- Interpretation: which feature explains more of the VIX, the trend $R_{1,t}$ or the historical volatility Σ_t ? Relate your answer to Facts 6 and 9 of Lecture 1
- Interpretation: if r^2 falls after May 2022, compare the RMSE and the variance of the VIX across periods; explain what this comparison shows and why it cannot, on its own, tell whether the relationship between the VIX and the S&P 500 path changed

Common Pitfalls

Data

- ▣ Mixing price and total-return indices
- ▣ Taking returns before aligning calendars in joint analyses
- ▣ Trusting quote series without checking extremes and autocorrelation
- ▣ Using 252 days for assets that trade every day

Statistics

- ▣ i.i.d. standard errors under volatility clustering
- ▣ Testing a risk model on the sample used to estimate it
- ▣ Reporting a kurtosis when the fourth moment may not exist
- ▣ Comparing Sharpe ratios without a test, or across currencies

- ▣ Rule: look at the data (charts, extremes, zero returns) before running any test

Take-Home Assignment (Team Project)

- Choose **three** assets from different markets (at least one listed on the Bucharest Stock Exchange)
- For each asset
 - ▶ Document the data: source, price vs total return, cleaning steps
 - ▶ Summary-statistics table of Lecture 1, with robust autocorrelation tests
 - ▶ Both tail indices with Hill plots; at least five stylised facts of [Cont \(2001\)](#)
- Compare risk-adjusted performance on a common sample, with a significance test
- Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary
- Crypto-assets: read [Härdle, Harvey & Reule \(2020\)](#) before choosing a coin
- Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

Code

- ▣ **Notebook:** `chapter1_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_01

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